Real-Time Cryptocurrency Volatility Detection

Project Scoping Brief

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1. Use Case: What Problem Are We Solving?

Business Context: Cryptocurrency markets are extremely volatile, with prices changing rapidly within seconds. Traders, investors, and automated trading systems need real-time alerts about sudden price volatility to make informed decisions about buying, selling, or adjusting positions.

The Problem:

- Bitcoin and other cryptocurrencies can experience price swings of 5-10% within minutes
- Current solutions rely on delayed data or manual monitoring
- Traders miss opportunities or incur losses due to delayed volatility detection
- No real-time system to predict short-term (60-second) volatility spikes

Who This Helps:

- Day traders making rapid trading decisions
- Automated trading bots needing volatility signals
- Risk management teams monitoring exposure
- Portfolio managers adjusting hedging strategies

Value Proposition: A real-time system that predicts volatility 60 seconds ahead, giving users time to:

- Execute trades before major price movements
- Adjust stop-loss orders to protect positions
- Reduce risk exposure during volatile periods
- Capitalize on arbitrage opportunities

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2. Prediction Goal: What Exactly Are We Predicting?

Primary Objective: Predict whether Bitcoin (BTC-USD) will experience HIGH or LOW volatility in the next 60 seconds.

Definition of Volatility:

- **High Volatility:** Price standard deviation > [threshold] over 60-second window
- **Low Volatility:** Price standard deviation ≤ [threshold] over 60-second window

Prediction Window:

- **Input:** Last 5 minutes of tick data (price, volume, bid-ask spread)

- **Output:** Binary classification (HIGH/LOW) for next 60 seconds
- **Update Frequency:** Every 10 seconds (rolling predictions)

Features We'll Use:

1. **Price-based:**
    - Rolling mean/std of prices (30s, 60s, 120s windows)
    - Price change velocity (rate of change)
    - High-low price range over recent windows
2. **Volume-based:**
    - 24-hour trading volume trends
    - Volume spikes (sudden increases)
    - Volume-price correlation
3. **Market microstructure:**
    - Bid-ask spread (market liquidity indicator)
    - Trade frequency (number of trades per second)
    - Buy vs sell pressure

Example: At 3:00:00 PM, using data from 2:55:00-3:00:00, predict if volatility at 3:01:00 will be HIGH or LOW.

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3. Success Metrics: How Do We Measure Performance?

Primary Metric: Accuracy

- Target: **≥ 70%** classification accuracy
- Rationale: Random guessing = 50%, industry standard for trading signals = 60-70%

Secondary Metrics:

1. **Precision (for HIGH volatility class)**
    - Target: **≥ 75%**
    - Why: False alarms are costly (unnecessary trades, fees)
    - When we predict HIGH, we want to be right 75% of the time
2. **Recall (for HIGH volatility class)**
    - Target: **≥ 65%**
    - Why: Missing volatile periods means missed opportunities
    - We want to catch at least 65% of actual high volatility events
3. **F1-Score**
    - Target: **≥ 0.70**
    - Why: Balance between precision and recall

4. **Prediction Latency**
    - Target: **< 100ms** from data arrival to prediction
    - Why: Real-time systems need fast responses
    - Measured: Time from Kafka message receipt to model output

Business Impact Metrics:

- **Alert Timeliness:** Predictions delivered 60 seconds before volatility event
- **System Uptime:** ≥ 99% availability during trading hours
- **Data Freshness:** ≤ 1 second delay from Coinbase to model input

Evaluation Approach:

- Train on 7 days of historical data
- Test on next 24 hours (hold-out set)
- Monitor performance drift over time with Evidently

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4. Risk Assumptions: What Could Go Wrong?

Technical Risks

1. **Data Availability**
    - **Risk:** Coinbase WebSocket disconnects or lags
    - **Mitigation:** Implemented reconnection logic with 5 retry attempts
    - **Impact if occurs:** Missing data → degraded predictions
    - **Monitoring:** Track connection uptime, message gaps
2. **Kafka Performance**
    - **Risk:** Kafka cannot handle message volume (>100 msgs/sec)
    - **Mitigation:** Kafka designed for high throughput; monitoring lag
    - **Impact if occurs:** Backpressure → prediction delays
    - **Monitoring:** Consumer lag metrics in Kafka
3. **Model Latency**
    - **Risk:** Predictions take >100ms, defeating "real-time" purpose
    - **Mitigation:** Use lightweight model (e.g., Random Forest, not deep learning)
    - **Impact if occurs:** Predictions arrive too late to be useful
    - **Monitoring:** Track inference time with MLflow

Data Quality Risks

4. **Concept Drift**
    - **Risk:** Market patterns change; model becomes stale
    - **Mitigation:** Evidently monitoring for data/prediction drift
    - **Impact if occurs:** Accuracy degrades over weeks/months
    - **Monitoring:** Weekly drift reports

5. **Insufficient Training Data**
    - **Risk:** 7 days may not capture all market conditions
    - **Assumption:** Recent data more relevant than old data for crypto
    - **Impact if occurs:** Model overfits to recent patterns
    - **Mitigation:** Cross-validation, monitor test performance

Business/Market Risks

6. **Market Regime Changes**
    - **Risk:** Bull/bear market transitions break patterns
    - **Assumption:** Volatility patterns are somewhat stable
    - **Impact if occurs:** Sudden accuracy drop during market crashes
    - **Mitigation:** Retrain model weekly; alert on performance drops
7. **Low Volatility Periods**
    - **Risk:** During stable markets, class imbalance (mostly LOW volatility)
    - **Assumption:** Need balanced training data
    - **Impact if occurs:** Model predicts LOW for everything
    - **Mitigation:** Class balancing techniques (SMOTE, class weights)
8. **Regulatory Changes**
    - **Risk:** Crypto regulations could affect market structure
    - **Assumption:** Coinbase API remains accessible and free
    - **Impact if occurs:** Need to switch data source
    - **Mitigation:** Design system to be data-source agnostic

Operational Risks

9. **System Failures**
    - **Risk:** Docker container crashes, service downtime
    - **Mitigation:** Docker restart policies, health checks
    - **Impact if occurs:** Missed trading opportunities
    - **Monitoring:** Uptime alerts, automated restarts
10. **Resource Constraints**
    - **Risk:** Running out of disk space (Kafka logs grow large)
    - **Assumption:** Adequate storage provisioned
    - **Impact if occurs:** System halts
    - **Mitigation:** Log rotation, storage monitoring

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5. Project Scope & Timeline

In Scope:

- Real-time data ingestion from Coinbase (BTC-USD)

- Kafka streaming infrastructure
- Feature engineering pipeline
- Binary volatility classification model
- MLflow experiment tracking
- Evidently drift monitoring
- Dockerized deployment

Out of Scope (for this project):

- Multi-currency predictions (only BTC-USD)
- Regression (exact volatility values)
- Trading execution/backtesting
- Production deployment to cloud
- Multi-model ensemble

Timeline:

- ✅ Milestone 1: Streaming setup (Week 1) - 64% complete
- 🔄 Milestone 2: Feature engineering, EDA (Week 2)
- 🔄 Milestone 3: Modeling, evaluation (Week 3)

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6. Success Criteria Summary

✅ **System is successful if:**

1. Achieves ≥70% accuracy on hold-out test set
2. Predictions delivered with <100ms latency
3. System runs continuously for 24+ hours without crashes
4. Drift monitoring detects and alerts on performance degradation
5. All code containerized and reproducible

📊 **Deliverables:**

- Working streaming pipeline (Kafka + WebSocket)
- Trained ML model with documented performance
- Drift monitoring dashboard
- Docker containers for all components
- Documentation and reproducible setup

Notes & Assumptions

- **Data Source:** Coinbase Pro WebSocket (free, no authentication required for market data)
- **Infrastructure:** Local Docker environment (sufficient for prototype)
- **Model Complexity:** Start simple (Random Forest), iterate if needed
- **Retraining Strategy:** Manual retraining weekly (not automated in this scope)
- **Target Users:** Educational/demonstration (not production trading system)

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